

If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

- a. All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.
- b. Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.
- c. All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.
- d. Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).
- e. Bonus problems are not required for credit, but they are worth extra credit.

Definitions. The following definitions will be used in the problems.

Definition 1 (1-Nearest Neighbor (1-NN) Classifier). Given a set of labeled points $\{(x_i, y_i)\}$ with $x_i \in \mathbb{R}^d$, the *1-nearest neighbor classifier* assigns a label to any query point $x \in \mathbb{R}^d$ by finding the closest training point in Euclidean distance:

$$\hat{y}(x) = y_j \quad \text{where } j = \arg \min_i \|x - x_i\|.$$

Definition 2 (Perpendicular Bisector Hyperplane). Given two points $a, b \in \mathbb{R}^d$, the perpendicular bisector hyperplane is the set of all points equidistant from a and b :

$$H = \{x \in \mathbb{R}^d : \|x - a\| = \|x - b\|\}.$$

Geometrically, H is the $(d - 1)$ -dimensional hyperplane that is perpendicular to the line segment joining a and b , and passes through the midpoint $\frac{a+b}{2}$.

Problems

Problem 1. (Nearest Neighbor Geometry.) Let $x \in \mathbb{R}^d$ and consider three points $a, b, c \in \mathbb{R}^d$.

- (a) Show that a is closer to x than b if and only if

$$(x - \frac{a+b}{2})^\top (a - b) < 0.$$

- (b) Conclude that the 1-NN decision boundary between a and b is the perpendicular bisector hyperplane.

Problem 2. (Weighted Norm Decision Boundaries.) In weighted 1-nearest neighbor (1-NN) classification, the decision boundary between two training points $a, b \in \mathbb{R}^d$ is defined as the set of points x such that

$$d_W(x, a) = d_W(x, b),$$

where the weighted Euclidean distance is

$$d_W(x, y) = \sqrt{(x - y)^\top W (x - y)}, \quad W = \text{diag}(\alpha_1, \dots, \alpha_d) = \begin{bmatrix} \alpha_1 & 0 & \cdots & 0 \\ 0 & \alpha_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_d \end{bmatrix}, \quad \alpha_i > 0.$$

- (a) Show that the decision boundary has the form of a hyperplane

$$c^\top x = d,$$

for some vector c and scalar d depending on a, b, W .

- (b) Specialize your result to $d = 2$ and write the equation of the decision boundary in terms of $\mathbb{R}^2$ coordinates x_1, x_2 .
- (c) Again with $d = 2$, show that if $\alpha_1 \neq \alpha_2$, this boundary is not perpendicular to the segment ab in the original coordinates.
- (c) Again with $d = 2$, let $z = Sx$ with $S = \text{diag}(\sqrt{\alpha_1}, \sqrt{\alpha_2})$. Show that in the z -coordinates the decision boundary is the usual perpendicular bisector of the transformed points Sa and Sb . Explain how the parameters (α_1, α_2) rescale the coordinate axes: if $\alpha_1 > \alpha_2$, distances along the x_1 direction are stretched more, so the first feature has greater influence in the weighted 1-NN classification, and similarly for $\alpha_2 > \alpha_1$.

Problem 3. (Norm Effects on k-Mean.)

Consider two clusters of points in $\mathbb{R}^2$: $A = \{(0, 0), (0, 2)\}$ and $B = \{(5, 0), (5, 2)\}$.

- (a) Compute the centroids of A and B .
- (b) For a new point $x = (2, 1)$, compute its distance to both centroids in the ℓ_2 and ℓ_1 norms.
- (c) Show that the assignment of x can differ depending on the choice of norm.

Problem 4. (1-NN Decision Boundaries under Different Norms.) Consider two training points $a = (0, 0)$ and $b = (2, 1)$ with different labels.

- (a) Under the ℓ_2 norm, write the equation of the 1-NN decision boundary (the set of x equidistant to a and b).
- (b) Under the ℓ_1 norm, write the equation of the 1-NN decision boundary. (Hint: solve $|x_1| + |x_2| = |x_1 - 2| + |x_2 - 1|$ by cases on the signs of the absolute values.)
- (c) Compare the two boundaries and explain the geometric difference.

Problem 5. (Monotonicity of k-Means Objective.) Show that updating centroids as the mean reduces the objective.

Problem 6. (Matrix View of k-Means.) For a matrix $A \in \mathbb{R}^{m \times n}$ with entries a_{ij} , the **Frobenius norm** is defined as

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}.$$

Equivalently, it is the Euclidean (ℓ_2) norm of the matrix viewed as a long vector of entries. Another useful identity is

$$\|A\|_F^2 = \text{trace}(A^\top A) \quad \text{where} \quad \text{trace}(M) = \sum_{i=1}^n m_{ii} \quad \text{for } M \in \mathbb{R}^{n \times n}.$$

Let $X \in \mathbb{R}^{n \times d}$ be the data matrix whose i th row is $x_i^\top$. Let $H \in \{0, 1\}^{n \times k}$ be the assignment matrix with exactly one 1 per row (i.e., row i has a 1 in column $r(i)$ if x_i is assigned to cluster $r(i)$). Let $C \in \mathbb{R}^{k \times d}$ be the centroid matrix whose j th row is $c_j^\top$. Show carefully that the k-means objective with fixed assignments H equals

$$\sum_{i=1}^n \|x_i - c_{r(i)}\|_2^2 = \|X - HC\|_F^2.$$

(Provide both a row-wise argument and, if you like, a trace/Frobenius argument.)

Problem 7. (Silhouette Score for k-Means.) Consider a clustering $\{C_1, \dots, C_k\}$ of n points in $\mathbb{R}^d$. When we write $C_j \setminus \{x_i\}$ we mean the set C_j with the point x_i removed. Recall from the lecture that the *silhouette score* of a point x_i in cluster C_j is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}},$$

where

$$a(i) = \frac{1}{|C_j| - 1} \sum_{x \in C_j \setminus \{x_i\}} \|x_i - x\|$$

is the average distance to other points in the same cluster, and

$$b(i) = \min_{q \in \{1, \dots, k\} \text{ s.t. } q \neq j} \frac{1}{|C_q|} \sum_{x \in C_q} \|x_i - x\|$$

is the average distance to the nearest other cluster; note here that $q \in \{1, \dots, k\}$ s.t. $q \neq j$ means that q is an index in the set $\{1, \dots, k\}$ that is not equal to j .

- Compute the silhouette score for each point in the clustering $C_1 = \{(0, 0), (1, 0)\}$, $C_2 = \{(5, 0), (6, 0)\}$.
- Show that $s(i) \in [-1, 1]$ and interpret the meaning of $s(i) \approx 1$, $s(i) \approx 0$, and $s(i) \approx -1$.
- Explain why maximizing the average silhouette score $\frac{1}{n} \sum_i s(i)$ is a reasonable clustering objective.

Problem 8. (Variance Decomposition and k -Means.) For a dataset $X = \{x_1, \dots, x_n\}$, let $\mu = \frac{1}{n} \sum_{i=1}^n x_i$ be the global mean and $\mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$ be the mean of cluster C_j .

- (a) Expand $\|u - v\|^2$ and show that

$$\|u - v\|^2 = \|u - w\|^2 + \|w - v\|^2 + 2(u - w)^\top (w - v).$$

- (b) Show that

$$\sum_{i=1}^n \|x_i - \mu\|^2 = \underbrace{\sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2}_{\text{within-cluster variance}} + \underbrace{\sum_{j=1}^k |C_j| \|\mu_j - \mu\|^2}_{\text{between-cluster variance}}.$$

This is interesting for the following reason: the total variance splits into within- and between-cluster parts. Minimizing the k -means objective (the within term) implicitly maximizes the between-cluster scatter at fixed total variance, promoting compact, separated clusters

Problem 9. (Linear Approximation of a Norm.) Recall that the first order Taylor approximation $\hat{f}$ of a differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ at x_0 is

$$\hat{f}(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0).$$

- Compute the gradient of the ℓ_2 norm $f(x) = \|x\|_2$ at a point x_0 with all nonzero entries.
- Write the first order Taylor approximation of $f(x) = \|x\|_2$ at a point x_0 with all nonzero entries.
- Repeat parts (a) and (b) for the case $f(x) = \|x\|_1$.

Bonus Problem 1. (Linear Approximation of a Norm.) For Problem 9 above, solve this extra bonus part. Both the ℓ_1 and the ℓ_2 norm are not differentiable at points with some zero entries. However, we can still write a "subgradient." Consider $f(x) = \|x\|_2$ and the case $x_0 = 0$. For a unit vector v , compute the directional derivative

$$\lim_{t \downarrow 0} \frac{\|tv\|_2 - \|0\|_2}{t}.$$

Does this limit depend on v ? What does this tell you about whether a single gradient exists at the origin?

Bonus Problem 2. (k -NN with Adversarial Perturbations.) Consider a 1-NN classifier trained on points $\{(x_i, y_i)\}_{i=1}^n$ in $\mathbb{R}^d$.

- Given a query point x with nearest neighbor x_j , find the *minimum* ℓ_2 perturbation δ such that $x + \delta$ has a different nearest neighbor. Express your answer in terms of the training points and x .
- Show that if training points are sparse (far apart), the classifier is more robust to perturbations.
- Design a simple 2D example where a perturbation of magnitude $\|\delta\|_2 = 0.1$ changes the classification from $+1$ to -1 .